

8

B

$$E_p = E_T$$

$$mgr = \frac{1}{2} m (4^2)$$

$$gr = 8$$

If the mass possesses 75% of E , it means E has decreased by 25%.

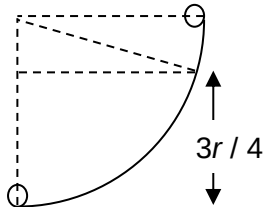
Let x be the point where GPE has decreased by $E/4$,

$$E_p = E_x$$

$$mgr = mg(3r/4) + \frac{1}{2} m v^2$$

$$v^2 = \frac{1}{2} gr$$

$$v = 2.0 \text{ m s}^{-1}$$



8

C

As $v = r\omega$